GFN

Microsoft Azure 5

Tag 13
08.07.2026
Mittwoch

Von KI erstellt, von Menschen kontrolliert.

Guten Morgen!

GitHub Repo https://github.com/www42/a5

Morgen: Mannheim

AZ-104

Learning Paths:

- ➡️ Manage identities and governance in Azure
- ➡️ Prerequisites for Azure administrators – CloudShell and templates
- ➡️ Configure and manage virtual networks for Azure administrators
- ➡️ Implement and manage storage in Azure
- ➡️ Deploy and manage Azure compute resources VM, Docker, Kubenetes, Function, Logic App
- ➡️ Monitor and back up Azure resources

Heute

- SAS , Rekeying (MeasureUp)
- Mini-Übung key Azure for Student
- Skillable Challange Lab

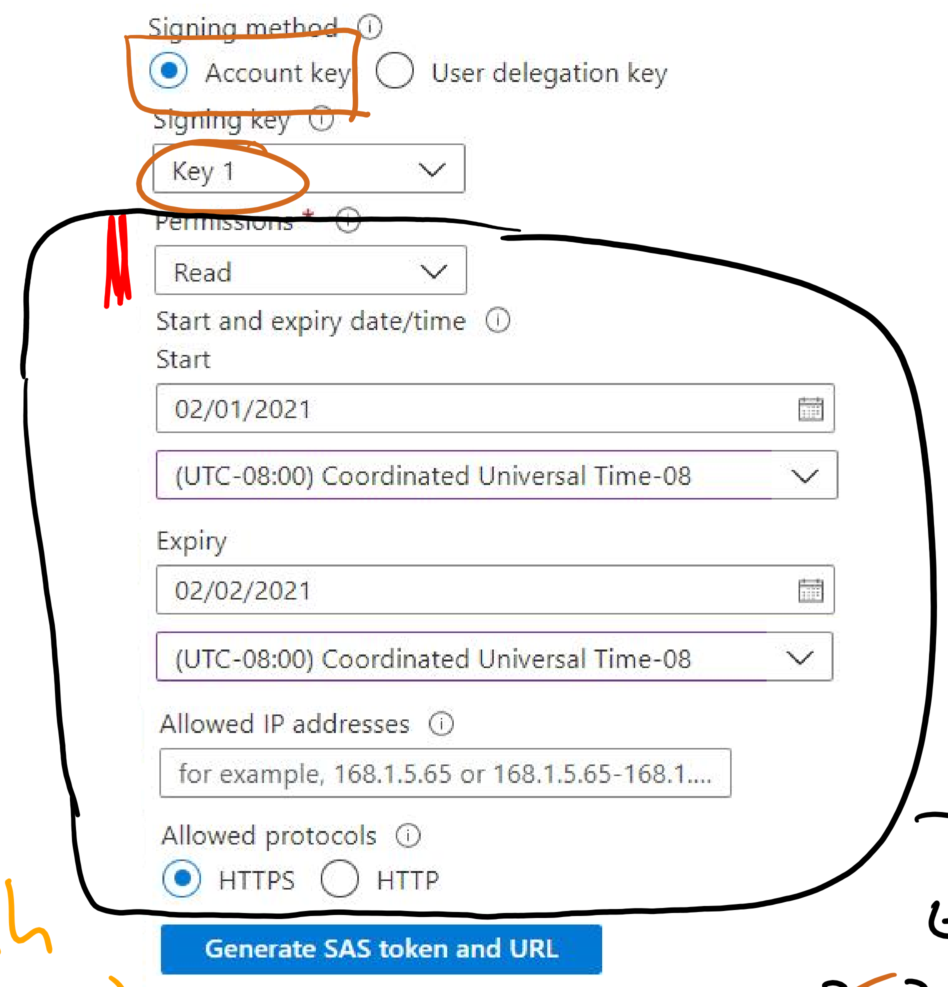
- Start LP Compute
- Create VM Portal Azure for Student
  Delete

What happened yesterday

SAS

Create Shared Access Signatures

- Provides delegated access to resources
- Grants access to clients without sharing your storage account keys
- The account SAS delegates access to resources in one or more of the storage services
- The service SAS delegates access to a resource in just one of the storage services



Was ist eine Signatur:

On key

Eine Signatur ist ein verschlüsselter Hash (= Püfsumme)

video.mp4 → SHA256 ~~1f7b28085...~~

verschl. On

MD5, SHA1, SHA2, SHA256 ...

SHA256

~~2x7abc~~

On key1

Mini-Übung

List and Rotate Storage Access Keys (Azure CLI)

Azure for Student

➡️ GitHub a5

Cloud Shell

Local Shell (VS Code) *

Schweizer

Erich Gamma

Eclipse

In this challenge, you will configure network peering between two Azure virtual networks. First, you will review the existing environment. Next, you will configure Azure virtual network peering. Finally, you will create two virtual machines by using Azure PowerShell, and then you will use the virtual machines to test the virtual network peering connections. Note: Once you begin the challenge, you will not be able to pause, save, or return to your challenge. Please ensure that you have set aside enough time to complete the challeng...

Available Self-Paced: Yes
Allow Multiple Attempts: Yes (Limit 5 attempts)

3

Implement an Azure Load Balancer [Guided]

Challenge Labs All Access Pass , AZ300.2-007

30 minutes

In this Challenge Lab, you will implement an Azure load balancer. First, you will create an Azure load balancer. Next, you will configure a backend pool for the load balancer, and then you will configure an HTTP health probe. Finally, you will create a load balancing rule. Note: Once you begin the challenge, you will not be able to pause, save, or return to your challenge. Please ensure that you have set aside enough time to complete the challenge before you start.

Additional Details 5

Required: No
Available Instructor-Led: Yes
Available Self-Paced: Yes
Allow Multiple Attempts: Yes (Limit 5 attempts)

8 Challenge Labs supporting AZ-104: Implement and manage storage in Azure

1

Manage Access to Azure Storage [Guided]

Challenge Labs All Access Pass , AZ300.1-002

30 minutes

In this Challenge Lab, you will secure an Azure storage account. First, you will configure a storage account to use a customer-managed storage service encryption key, and then you will configure the storage account to use secure transfer. Next, you will configure firewall rules to restrict storage account access to a specific IP address, and then you will generate a shared access signature to be used to grant time-limited, delegated access to a storage account. Finally, you will test the access to the storage account by using...

Additional Details 5

Required: No
Available Instructor-Led: Yes
Available Self-Paced: Yes
Allow Multiple Attempts: Yes (Limit 5 attempts)

9 Challenge Labs supporting AZ-104: Deploy and manage Azure compute resources

1

Enable Azure Virtual Machine Scale Sets for High Availability and Scalability [Guided]

Challenge Labs All Access Pass , HCS-009

30 minutes

In this challenge, you will create and deploy multiple Azure virtual machine scale sets acting as web server and app server tiers. First, you will create a virtual machine scale set for the web server tier. Next, you will use an Azure Resource Manager (ARM) template to create a virtual machine scale set for the app server tier. Finally, you will verify connectivity to the virtual machine scale set for the web server tier. Note: Once you begin a challenge you will not be able to pause, save, or return to your progress. Please ensure you have set...

Additional Details 5

Required: No
Available Instructor-Led: Yes
Available Self-Paced: Yes
Allow Multiple Attempts: Yes (Limit 5 attempts)

10 Challenge Labs supporting AZ-104: Monitor and back up Azure resources

1

Implement Azure Backup for Azure Virtual Machines [Guided]

Challenge Labs All Access Pass , AZ300.2-009

30 minutes

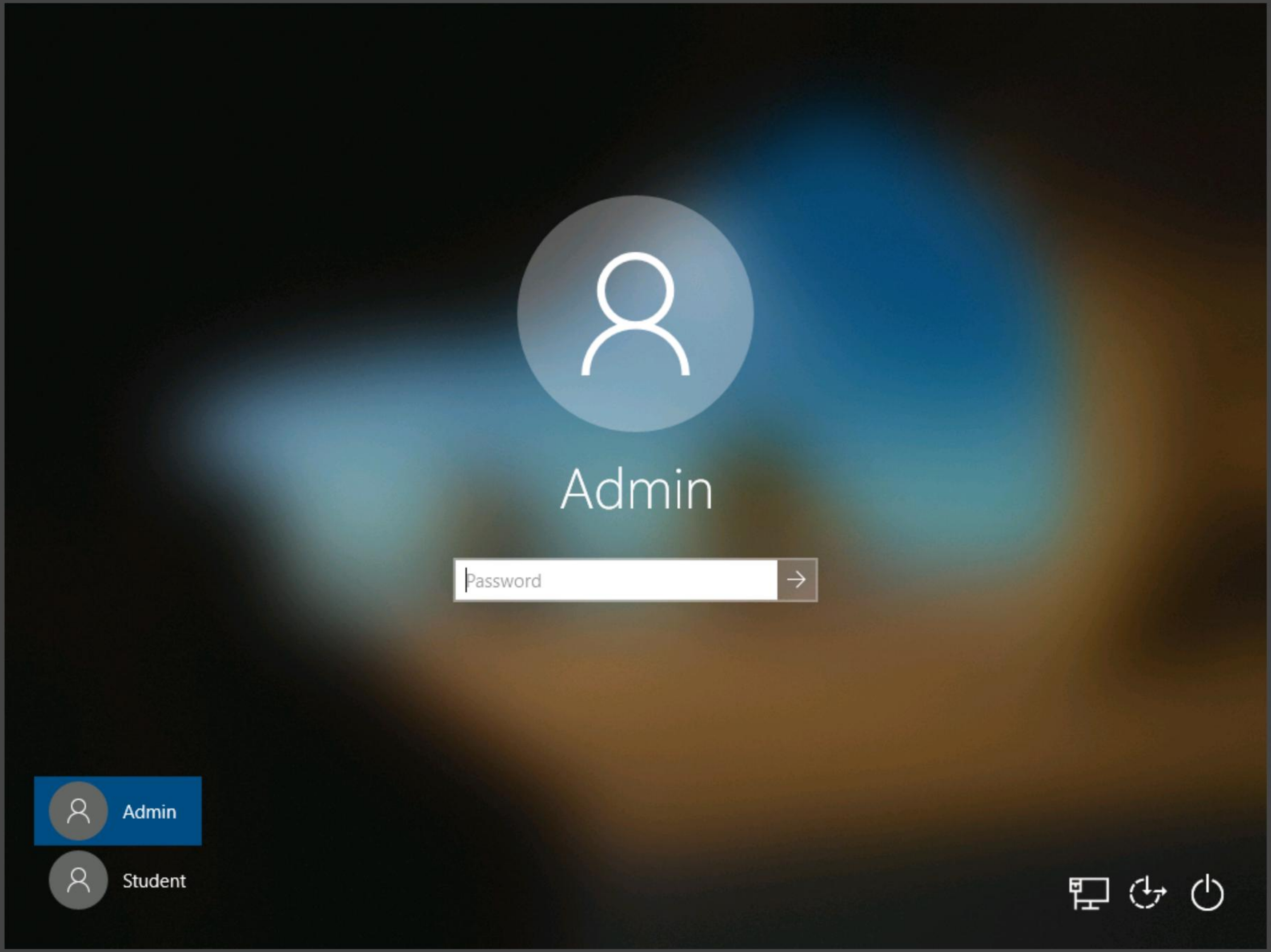
In this challenge, you will implement Azure backup for your Azure virtual machines. First, you will back up an Azure virtual machine by using the Azure portal. Next, you will enable

Additional Details 5

Required: No
Available Instructor-Led: Yes
Available Self-Paced: Yes

AZ300.1-002: Manage Access to Azure Storage [Guided] - Personal - Microsoft Edge

https://labclient.labondemand.com/LabClient/bb8be07e-4fda-4439-8a85-e4441401a1ed



AZ300.1-002: Manage Access to Azure Storage [Guided]

30 Minutes Remaining

Instructions Resources Help 100%

skillable challenges

Guided

Manage Access to Azure Storage

Challenge Overview

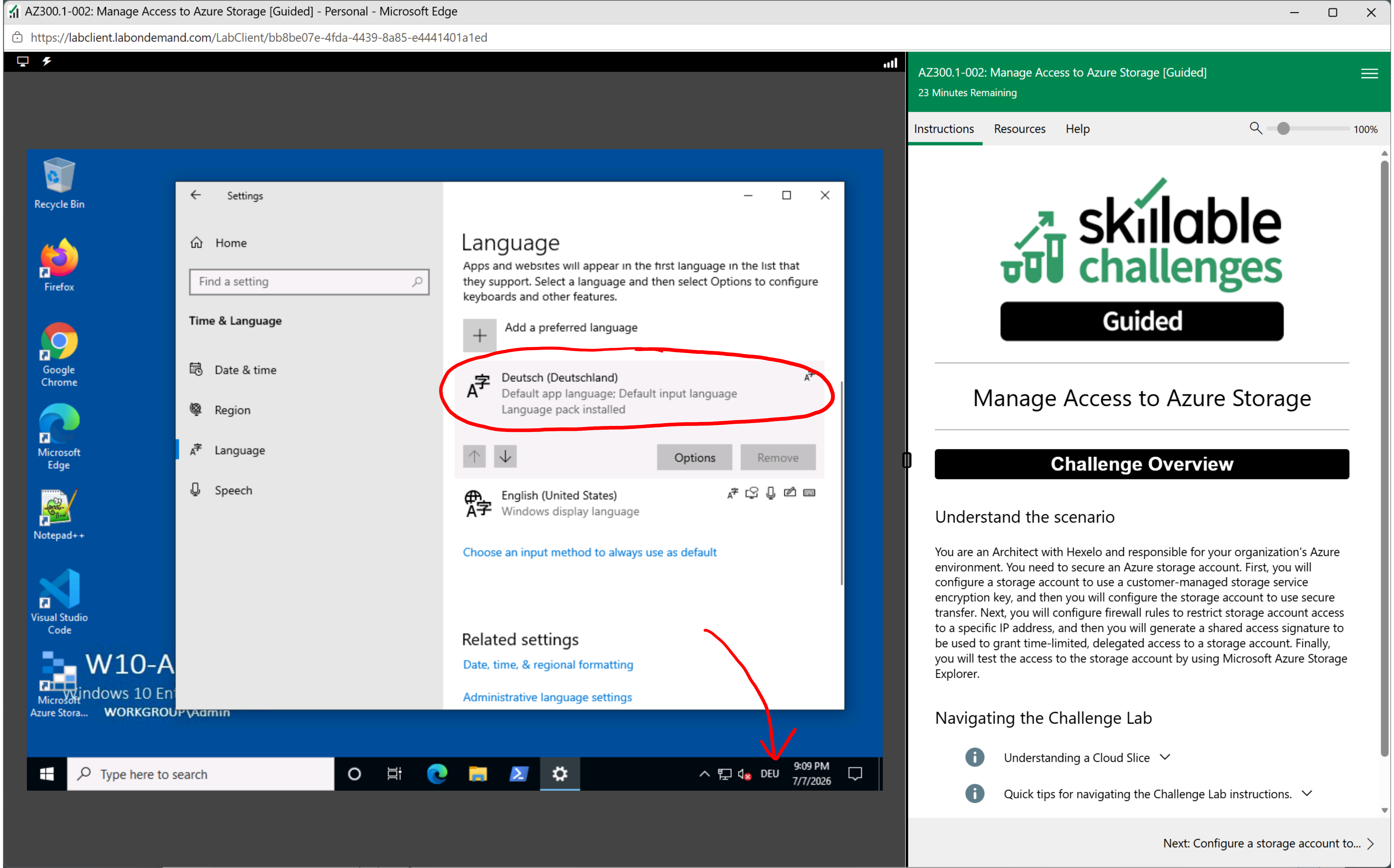
Understand the scenario

You are an Architect with Hexelo and responsible for your organization's Azure environment. You need to secure an Azure storage account. First, you will configure a storage account to use a customer-managed storage service encryption key, and then you will configure the storage account to use secure transfer. Next, you will configure firewall rules to restrict storage account access to a specific IP address, and then you will generate a shared access signature to be used to grant time-limited, delegated access to a storage account. Finally, you will test the access to the storage account by using Microsoft Azure Storage Explorer.

Navigating the Challenge Lab

- Understanding a Cloud Slice
- Quick tips for navigating the Challenge Lab instructions.

Next: Configure a storage account to... >


AZ300.1-002: Manage Access to Azure Storage [Guided] - Personal - Microsoft Edge
https://labclient.labondemand.com/LabClient/bb8be07e-4fda-4439-8a85-e4441401a1ed
Recycle Bin
Firefox
Google Chrome
Microsoft Edge
Notepad++
Visual Studio Code
Microsoft Azure Stora...
W10-A
Windows 10 En
WORKGROUP\Admin
Settings
Home
Find a setting
Time & Language
Date & time
Region
Language
Speech
Language
Apps and websites will appear in the first language in the list that they support. Select a language and then select Options to configure keyboards and other features.
Add a preferred language
Deutsch (Deutschland)
Default app language; Default input language
Language pack installed
Options
Remove
English (United States)
Windows display language
Choose an input method to always use as default
Related settings
Date, time, & regional formatting
Administrative language settings
Type here to search
DEU
9:09 PM
7/7/2026
AZ300.1-002: Manage Access to Azure Storage [Guided]
23 Minutes Remaining
Instructions
Resources
Help
100%
skillable challenges
Guided
Manage Access to Azure Storage
Challenge Overview
Understand the scenario
You are an Architect with Hexelo and responsible for your organization's Azure environment. You need to secure an Azure storage account. First, you will configure a storage account to use a customer-managed storage service encryption key, and then you will configure the storage account to use secure transfer. Next, you will configure firewall rules to restrict storage account access to a specific IP address, and then you will generate a shared access signature to be used to grant time-limited, delegated access to a storage account. Finally, you will test the access to the storage account by using Microsoft Azure Storage Explorer.
Navigating the Challenge Lab
Understanding a Cloud Slice
Quick tips for navigating the Challenge Lab instructions.
Next: Configure a storage account to...

Microsoft

AZ-104

➡️ Deploy and manage Azure compute resources

VM

Vorteile

gibts schon lange

VMWare

Microsoft: Hyper-V

Kapselung

Nachteile

Kosten (in Azure)

Resourcen intensiv

Lange Boot zeiten

Docker Container

Introduction to Azure Virtual Machines

Scalierbarkeit

VM Scale Set (VMSS)

Snowflake Server

App Service

VM

NIST paper

Review Cloud Services Responsibilities

	Responsibility	SaaS	PaaS	IaaS	On-prem
Responsibility always retained by the customer	Information and data	Customer	Customer	Customer	Customer
	Devices (Mobile and PCs)	Customer	Customer	Customer	Customer
	Accounts and identities	Customer	Customer	Customer	Customer
Responsibility varies by type	Identity and directory infrastructure	Shared	Shared	Customer	Customer
	Applications	Microsoft	Shared	Customer	Customer
	Network controls	Microsoft	Shared	Customer	Customer
	Operating system	Microsoft	Microsoft	Customer	Customer
Responsibility transfers to cloud provider	Physical hosts	Microsoft	Microsoft	Microsoft	Customer
	Physical network	Microsoft	Microsoft	Microsoft	Customer
	Physical datacenter	Microsoft	Microsoft	Microsoft	Customer

Microsoft Customer Shared

Connection

NSG 3389

VM

Disk

NetworkInterface

VNet

Region

Azure for Students
5 Regions

Plan Virtual Machines

- Start with the network
- Name the virtual machine
- Choose a location = Region
  - Each region has different hardware and service capabilities
  - Locate Virtual Machines as close as possible to your users and to ensure compliance and legal obligations
- Consider pricing

Compute Size
+
Disk

Standard_DS2v3

Besser: Bastion Host

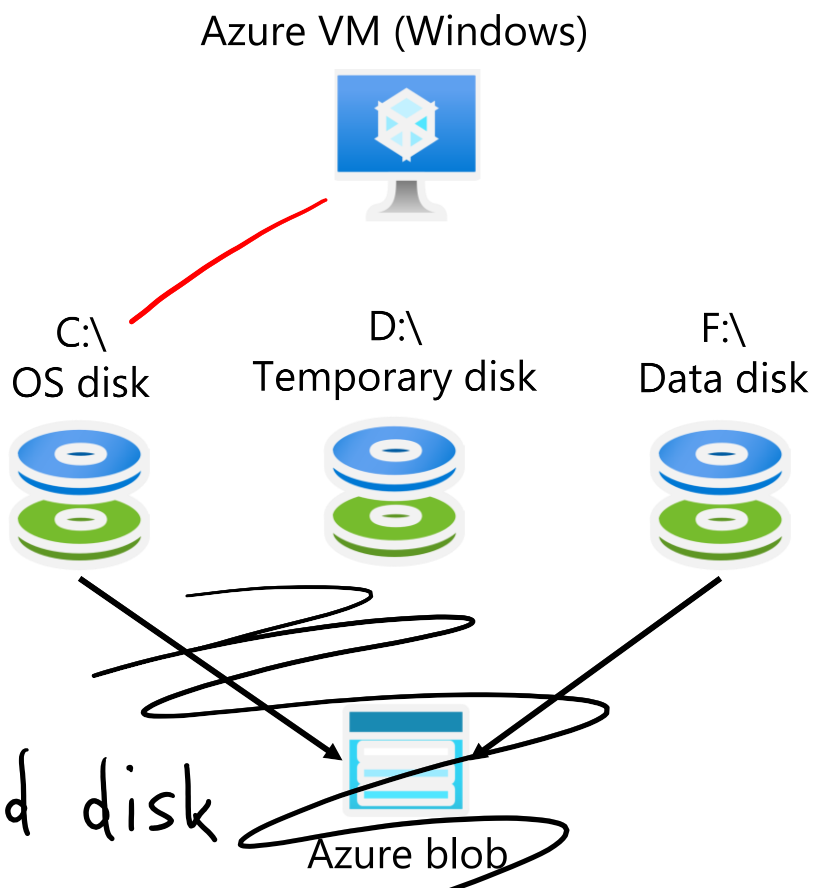
Determine Virtual Machine Sizing

Family Type	Description
General purpose B2	Balanced CPU-to-memory ratio.
Compute optimized	High CPU-to-memory ratio.
Memory optimized	High memory-to-CPU ratio.
Storage optimized	High disk throughput and I/O.
GPU	Specialized virtual machines targeted for heavy graphic rendering and video editing.
High performance compute	Our fastest and most powerful CPU virtual machines

(Lab 03 ? 5 Disks)

managed disk

Determine Virtual Machine Storage



Each Azure VM has two or more disks:

- OS disk
- Temporary disk (not all SKUs have one, content can be lost)
- Data disks (optional)

OS and data disks reside in Azure Storage accounts:

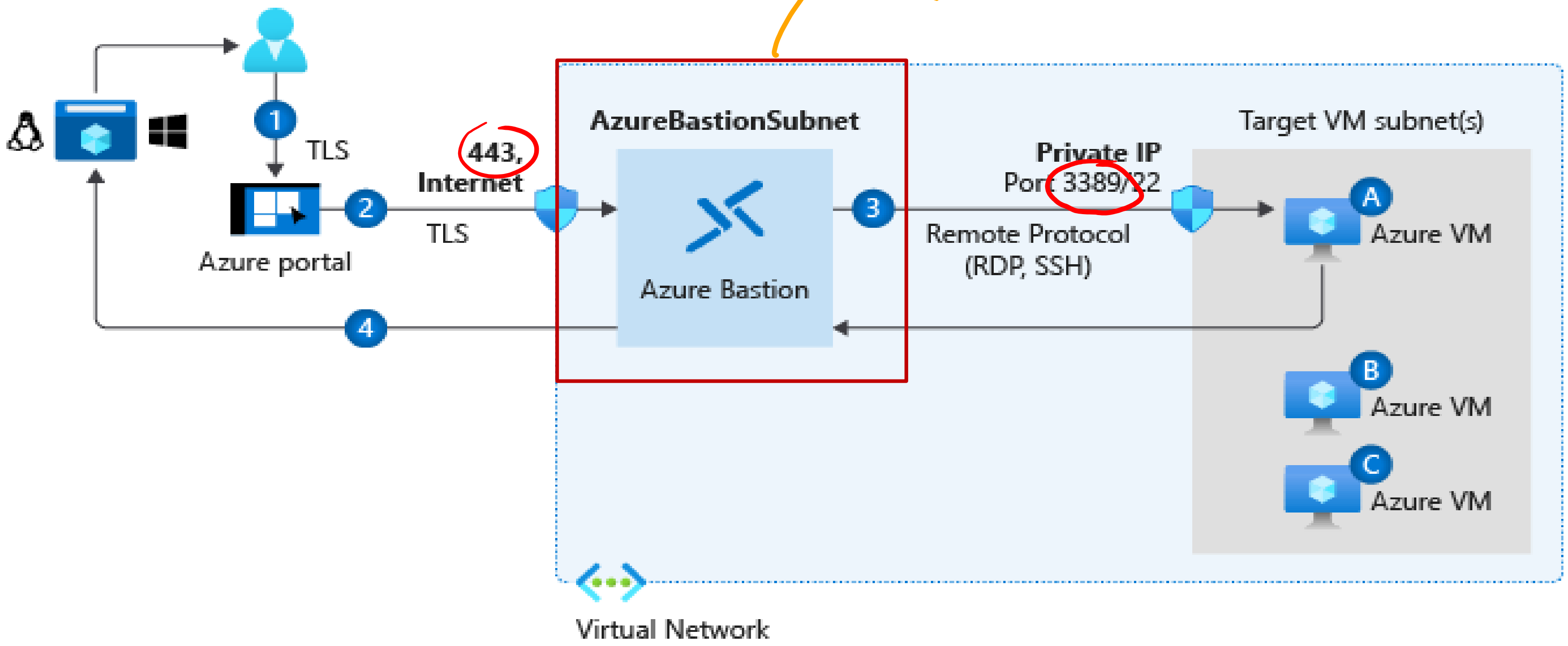
- Azure-based storage service
- Standard (HDD, SSD) or Premium (SSD), or Ultra (SSD)

Azure VMs use managed disks

unmanaged disk

Connect to Virtual Machines

Student SKU (Free)
(Developer?)



Bastion Subnet for RDP/SSH through the Portal over SSL

Remote Desktop Protocol for Windows-based Virtual Machines

Secure Shell Protocol for Linux based Virtual Machines

Labs


TO THE LAB ZONE →
NO THROUGH ROUTE
Berlin
Europa
Dublin
SPEEDY ROUTE
4²
tech·ed
lounge
software diagnostics
E 94
SD Studio
for Team Foundation Server
Visualize your code in
a software map.
PREWIS
software diagnostics
DataCore

Azure for Students

1) VM Deployment

Region
Windows Server
Size ? B2S...
ARM ? x86!
VNet
NSG

2) Connect to VM

Bastion Host (Free Tier)

3) Create Web site

wenn Windows → IIS

4) Advanced Settings

5) Alles löschen

End of presentation